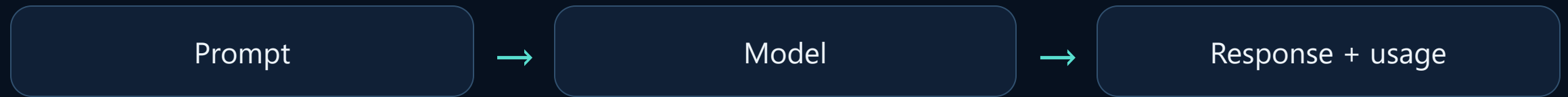


Meet the LLM.

Understand what goes in, what comes back, and how to ask for something useful.



TODAY'S
OUTCOME

Explain it. Run it. Record the evidence.

Understand

Plain words

Model, token, context,
prompt, request, response,
hallucination.

Build

Repeatable call

Run one script and inspect
the complete reply.

Prove

Prompt notes

Record behavior, usage,
variation, and limits.

**THE REAL
PROBLEM**

**“It looked good”
is not enough.**

Useful task

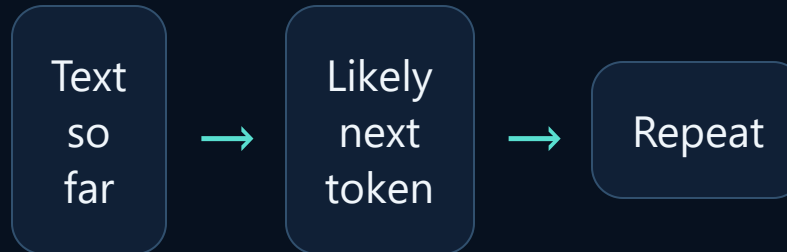
Draft three friendly tips for a new employee's first day in IT.

Team questions

- What did we send?
- What came back?
- What did it use?
- What still needs checking?

MENTAL
MODEL

An LLM predicts likely next tokens.



Powerful language
generation does not
equal guaranteed truth.

**CRITICAL
LIMIT**

Confident language can still be wrong.

Fluent

The response can be clear, polished, and believable.

False

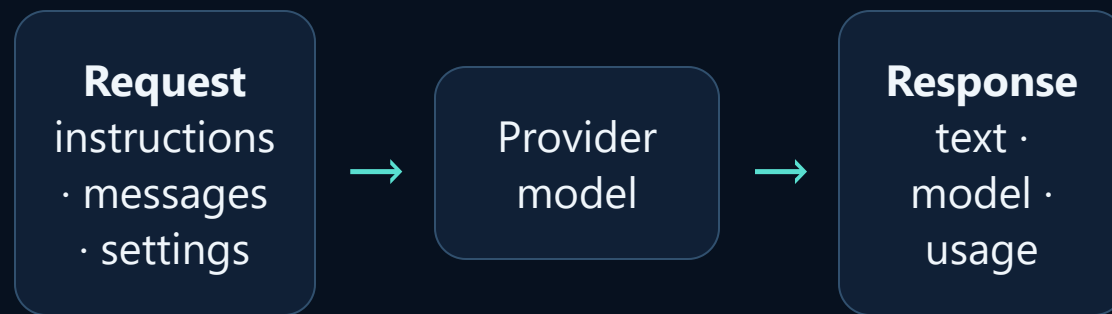
It can invent a fact, source, date, or explanation.

Rule:

confidence is not evidence.

SYSTEM
BOUNDARY

A model call has two sides.



TOKENS

Tokens are the pieces the model processes.

Input

Instructions,
messages,
context

Output

Generated response

Usage

Limits and usually
cost

CONTEXT
WINDOW

**The model has
limited
working space.**

Rules

History

Everything needed for this answer must fit. Context is not permanent memory.

Documents

Question

Close the important gaps.

1

Task

What should it do?

2

Input

What should it work with?

3

Constraints

Audience, length, tone, rules.

4

Format

Bullets, table, JSON, prose.

BEFORE
AND
AFTER

Make the result checkable.

Weak

Tips for a new job.

Topic, amount, tone, and shape are open.

Clear

Give exactly three friendly tips
for a first day at a new IT job.
Use one short bullet per tip.

We can check each condition.

FINISHED
RESULT

Answer text plus the meter.

```
--- The AI's answer ---  
[generated tips]
```

```
--- The meter (tokens &  
cost) ---  
Model: [configured model]  
Tokens: [input] sent in,  
[output] came back  
Approx cost: [calculated  
estimate]
```

Illustrative shape. Live
wording and numbers vary.

A message has a role and content.

```
messages = [  
    {  
        "role": "user",  
        "content": "Give me three  
friendly tips ..."  
    }  
]  
  
reply = chat(messages)
```

The reply is more than text.

.text

Generated answer

.model

Configured model

.input_tokens

Request usage

.output_tokens

Response usage

.cost_usd is a local
estimate, not an
invoice.

LIVE
DEMO

Run the real request.

```
python phase-1-  
foundations/day-01/labs/  
lab-1.1-first-ai-  
call/solution/first_call.py
```

Success

Meaningful text and
usage fields appear.

Failure

Read the first useful
error. Name the layer.

REPEATABILITY

**Same code does
not promise
identical words.**



Change the prompt. Observe the result.

A

Vague

B

Specific +
format

C1

Tricky
question

C2

Careful
system rule

SYSTEM
INSTRUCTION

Set the job and behavior.

You are a careful,
honest assistant.
If you are not sure,
say so.
Never invent facts.

A useful rule can reduce risk. It
cannot guarantee truth.

EVIDENCE

Record observations, not impressions.

Intended

Topic, audience,
length, tone, format.

Observed

What followed?
What changed?

Limits

What does one run
not prove?

TROUBLESHOOTING

Find the failing layer.

Local

Python · venv ·
import

Access

Key · account ·
billing

Outside

Network ·
provider · rate
limit

Read → name the layer →
change one thing → rerun.

COMPLETION
GATE

Day 1 is complete when evidence exists.

- ✓ First-call script returns text and usage.
- ✓ Prompt comparison has been observed.
- ✓ Prompt notes record behavior and limits.
- ✓ No secret or private data was exposed.
- ✓ You can explain the seven key terms.

Next: reliable outputs that software can validate.